

CHAPTER

CONTENTS

PHASE RULE
WHAT IS MEANT BY A 'PHASE'?
WHAT IS MEANT BY 'COMPONENTS'?
DEGREES OF FREEDOM
DERIVATION OF THE PHASE RULE
ONE-COMPONENT SYSTEM
PHASE DIAGRAMS
POLYMORPHISM
EXPERIMENTAL DETERMINATION OF TRANSITION POINT
THE WATER SYSTEM
THE SULPHUR SYSTEM
TWO-COMPONENT SYSTEMS
THE SILVER-LEAD SYSTEM
THE ZINC-CADMIUM SYSTEM
THE POTASSIUM DIOXIDE-WATER SYSTEM
THE MAGNESIUM-ZINC SYSTEM
THE FERRIC CHLORIDE-WATER SYSTEM
THE SODIUM SULPHATE-WATER SYSTEM



THE STATEMENT

The phase Rule is an important generalization dealing with the behaviour of heterogeneous systems. In general it may be said that with the application of *phase rule* it is possible to predict qualitatively by means of a diagram the effect of changing pressure, temperature and concentration on a heterogeneous system in equilibrium. This relationship governing all heterogeneous equilibria was first discovered as early as 1874 by an American physicist Willard Gibbs. **Gibb's Phase Rule** is free from flaws and limitations which are a common feature of all other generalizations of Physical Chemistry based on hypothetical assumptions as to the nature of the constitution of matter. It may be stated mathematically as follows :

$$F = C - P + 2$$

where F is the number of *degrees of freedom*, C is the number of *components* and P is the number of *phases* of the system. The terms 'phase', 'component', and 'degree of freedom' involved in the statement of the Phase Rule have a special significance and a clear understanding of these terms is essential before we proceed further with the subject.

WHAT IS MEANT BY A 'PHASE' ?

A phase may be defined as : **any homogeneous part of a system having all physical and chemical properties the same throughout.** A system may consist of one phase or more than one phases.

- (1) A system containing only liquid water is **one-phase** or **1-phase system** ($P = 1$)
- (2) A system containing liquid water and water vapour (a gas) is a **two-phase** or **2-phase system** ($P = 2$).
- (3) A system containing liquid water, water vapour and solid ice is a **three-phase** or **3-phase system**.

A system consisting of one phase only is called a **homogeneous system**.

A system consisting of two or more phases is called a **heterogeneous system**.

Explanation of the Term 'Phase'

Ordinarily three states of matter—gas, liquid, and solid are known as phases. However in *phase rule*, a **uniform part of a system in equilibrium is termed a 'phase'**. Thus a liquid or a solid mixture could have two or more phases.

Let us consider a few examples to understand the meaning of the term phase as encountered in phase rule.

(1) **Pure substances.** A pure substance (solid, liquid, or gas) made of one chemical species only, is considered as one phase. Thus oxygen (O_2), benzene (C_6H_6), and ice (H_2O) are all 1-phase systems. *It must be remembered that a phase may or may not be continuous.* Thus, whether ice is present in one block or many pieces, it is considered one phase.

(2) **Mixtures of gases.** All gases mix freely to form homogeneous mixtures. Therefore any mixture of gases, say O_2 and N_2 , is a 1-phase system.

(3) **Miscible liquids.** Two completely miscible liquids yield a uniform solution. Thus a solution of ethanol and water is a 1-phase system.

(4) **Non-miscible liquids.** A mixture of two non-miscible liquids on standing forms two separate layers. Hence a mixture of chloroform ($CHCl_3$) and water constitutes a 2-phase system.

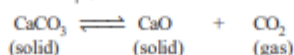
(5) **Aqueous solutions.** An aqueous solution of a solid substance such as sodium chloride (or sugar) is uniform throughout. Therefore it is a 1-phase system.

However, a saturated solution of sodium chloride in contact with excess solid sodium chloride is a 2-phase system.

(6) **Mixtures of solids.** (i) By definition, a **phase must have throughout the same physical and chemical properties**. Ordinary sulphur as it occurs in nature is a mixture of monoclinic and rhombic sulphur. These allotropes of sulphur consist of the same chemical species but differ in physical properties. Thus mixture of two allotropes is a 2-phase system.

(ii) **A mixture of two or more chemical substances contains as many phases.** Each of these substances having different physical and chemical properties makes a separate phase. Thus a mixture of calcium carbonate ($CaCO_3$) and calcium oxide (CaO) constitutes two phases.

Let us consider the equilibrium system : the **Decomposition of Calcium carbonate**. When calcium carbonate is heated in a closed vessel, we have



There are two solid phases and one gas phase. Hence it is a 3-phase system.

WHAT IS MEANT BY 'COMPONENTS'?

A system ' C ' in the *Phase Rule equation* stands for the number of components of a system in equilibrium. The term component may be defined as : **the least number of independent chemical constituents in terms of which the composition of every phase can be expressed by means of a chemical equation.**

Explanation of the Term 'Component'

To understand the above definition and to use it for finding the number of components of a system, remember that :

- The chemical formula representing the composition of a phase is written on LHS.
- The rest of the chemical constituents existing independently in the system as represented by chemical formulas are placed on RHS.
- The quantities of constituents on RHS can be made minus (-) or zero (0) to get at the composition of the phase on LHS.

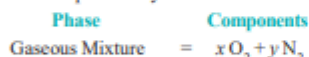
Now let us proceed to discuss the number of components of some systems.

(1) **Water and sulphur systems are 1-component systems.** Water system has three phases : *ice, water, and water vapour*. The composition of all the three phases is expressed in terms of one chemical individual H_2O . Thus water system has one component only.

Sulphur system has four phases : rhombic sulphur, monoclinic sulphur, liquid sulphur and sulphur vapour. The composition of all these phases can be expressed by one chemical individual sulphur (S). Hence it is a 1-component system.

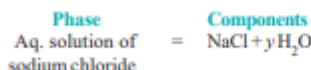
As is clear from above, when all the phases of a system can be expressed in terms of one chemical individual, it is designated as a one-component or 1-component system.

(2) **Mixture of gases.** A mixture of gases, say O_2 and N_2 , constitutes one phase only. Its composition can be expressed by two chemical substances O_2 and N_2 .



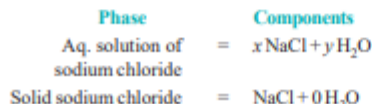
Hence a mixture of O_2 and N_2 has two components. In general the number of components of a gaseous mixture is given by the number of individual gases present.

(3) **Sodium chloride solution.** A solution of sodium chloride in water is a 1-phase system. Its composition ($xNaCl.yH_2O$) can be expressed in terms of two chemical individuals, sodium chloride and water.



Therefore an aqueous solution of sodium chloride or any other solute is a two-component or 2-component system.

A saturated solution of sodium chloride, in contact with excess solid sodium chloride has two phase, namely aqueous solution and solid sodium chloride. The composition of both phase can be expressed in terms of two chemical individuals $NaCl$ and H_2O .



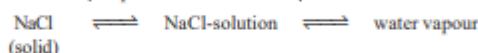
Hence a saturated solution of sodium chloride or any other solute in contact with solid solute, is 2-component system.

(1) **For a pure gas, $F = 2$.** For a given sample of any pure gas $PV = RT$. If the values of pressure (P) and temperature (T) be specified, volume (V) can have only one definite value, or that the volume (the third variable) is fixed automatically. Any other sample of the gas under the same pressure and temperature as specified above, will be identical with the first one. Hence a **system containing a pure gas has two degrees of freedom ($F = 2$)**.

(2) **For a mixture of gases, $F = 3$.** A system containing a mixture of two or more gases is completely defined when its composition, temperature and pressure are specified. If pressure and temperature only are specified, the third variable *i.e.*, composition could be varied. Since it is necessary to specify three variables to define the system completely, a **mixture of gases has three degrees of freedom ($F = 3$)**.

(3) **For water $\rightleftharpoons$ water vapour, $F = 1$.** The system water in equilibrium with water vapour, has two variables *temperature and pressure*. At a definite temperature the vapour pressure of water can have only one fixed value. Thus if one variable (temperature or pressure) is specified, the other is fixed automatically. Hence **the system water has one degree of freedom ($F = 1$)**.

(4) **For saturated NaCl solution, $F = 1$.** The saturated solution of sodium chloride in equilibrium with solid sodium chloride and water vapour.



Thus the system is completely defined if we specify temperature only. The other two variables *i.e.*, the composition of NaCl-solution (solubility) and vapour pressure have a definite value at a fixed temperature. Hence **the system has one degree of freedom**.

(5) **For ice-water-vapour system, $F = 0$.** In the system $\text{ice} \rightleftharpoons \text{water} \rightleftharpoons \text{vapour}$, the three phases coexist at the freezing point of water. Since the freezing temperature of water has a fixed value, the vapour pressure of water has also a definite value. The system has two variables (temperature and pressure) and both these are already fixed. Thus the system is completely defined automatically, there being no need to specify any variable. Hence **it has no degree of freedom ($F = 0$)**.

DERIVATION OF THE PHASE RULE

Consider a heterogeneous system in equilibrium of C components in which P phases are present. We have to determine the degrees of freedom of this system *i.e.*, the number of variables which must be arbitrarily fixed in order to define the system completely. Since the state of the system will depend upon the temperature and the pressure, these *two variables* are always there. The concentration variables, however, depend upon the number of phases. In order to define the composition of each phase it is necessary to specify the concentration of $(C - 1)$ constituents of each phase, the concentration of the remaining component being determined by difference. For P phases, therefore, the total number of concentration variables will be $P(C - 1)$ and these along with the two variables mentioned above *viz.*, temperature and pressure, make the total number of the variables of the system equal to $[P(C - 1) + 2]$.

On thermodynamic consideration when a system is in equilibrium, the partial molal free energy of each constituent of a phase is equal to the partial molal free energy of the same constituent in every other phase. Since the partial molal free energy of the constituents of a phase is a function of the temperature, pressure and $(C - 1)$ concentration variables, it follows that if there is one component in two phases, it is possible to write one equation amongst the variables and if there is one component in three phases, this fact may be written with the help of *two equations*. In general, therefore, when P phases are present, $(P - 1)$ equations are available for each component and for C components, the total number of equations or variables are $C(P - 1)$.

Since the number of equations is equal to the number of variables, the number of *unknown*

variables or degrees of freedom (F) will be :

$$\begin{aligned} F &= \text{No. of variables} - \text{Number of equations} \\ &= [P(C-1) + 2] - [C(P-1)] \\ &= PC - P + 2 - PC + C \\ &= C - P + 2 \\ F &= C - P + 2 \end{aligned}$$

ONE-COMPONENT SYSTEM

For a one-component system we can write the phase rule equation as :

$$F = C - P + 2 = 1 - P + 2 = 3 - P$$

Three cases may arise :

Case 1. When only **one phase** is present,

$$\therefore F = 3 - 1 = 2$$

Thus the system is **bivariant**. It can be completely defined by specifying the two variables, temperature and pressure. Or that, both the temperature and pressure can be varied independently. **Therefore a single phase is represented by an area on P, T-graph.**

Case 2. When **two phases** are in equilibrium,

$$F = 3 - 2 = 1$$

The system then has one degree of freedom and is termed **monovariant**. This means that the pressure cannot be changed independently if we change the temperature. The pressure is fixed automatically for a given temperature. **A two-phase system is depicted by a line on a P, T-graph.**

Case 3. When **three phases** are in equilibrium,

$$F = 3 - P = 3 - 3 = 0$$

$$\therefore F = 0$$

The system has zero degree of freedom and is termed **nonvariant or invariant**. This special condition can be attained at a definite temperature and pressure. The system is, therefore, defined completely and no further statement of external conditions is necessary. **A three-phase system is depicted by a point on the P, T-graph.** At this point the three phases (solid, liquid, vapour) are in equilibrium and, therefore, it is referred to as the **Triple point**.

PHASE DIAGRAMS

A phase diagram is a **plot showing the conditions of pressure and temperature under which two or more physical states can exist together in a state of dynamic equilibrium**. Fig. 19.1 is a typical phase diagram for a one-component system. The diagram consists of : (a) the Regions or Areas; (b) the Lines or Curves; and (c) the Triple point.

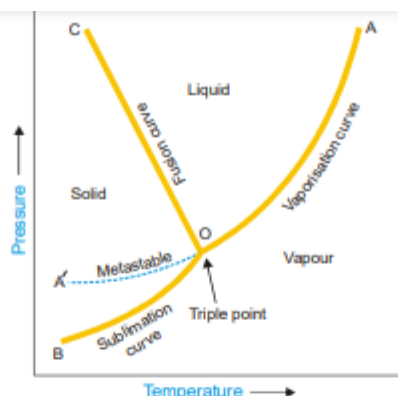
The significance of the characteristic features (Areas, Curves, Triple point) of a phase diagram are discussed below.

(1) Regions or Areas

The diagram is divided into three regions or areas which are labelled as 'solid', 'liquid', and 'vapour'. These areas in Fig. 19.1 are *COB*, *COA* and *AOB*. **Each of the three areas shows the conditions of temperature and pressure under which the respective phase can exist.** Applying the phase rule to the system when only one phase is present, we have

$$F = 1 - 1 + 2 = 2$$

i.e., each single phase has two degrees of freedom.



■ **Figure 19.1**
A typical phase diagram of a one component system.

Thus each area of phase diagram represents a bivariant system. At conditions wholly within one of the three areas, both the variables (pressure and temperature) can be varied independently without causing a change in the state of equilibrium. Therefore in order to define the condition of the phase both pressure and temperature must be stated.

(2) Lines or Curves

There are three lines or curves separating the regions or areas. These **curves show the conditions of equilibrium between any two of the three phases** i.e., solid/liquid, liquid/vapour, solid/vapour.

- Solid/liquid line (OC) which represents the equilibrium $Solid \rightleftharpoons Liquid$, is referred to as the **Melting curve** or **Fusion curve**.
- Liquid/vapour line (OA) which represents the equilibrium $Liquid \rightleftharpoons Vapour$, is referred to as the **Vapour Pressure curve** or **Vaporisation curve** for the liquid.
- Solid/vapour line (OB), which represents the equilibrium $Solid \rightleftharpoons Vapour$, is referred to as the **Sublimation curve**.

Applying phase rule to a one-component two-phase system.

$$F = C - P + 2 = 1 - 2 + 2 = 1$$

Thus phase rule predicts that the two phase equilibria stated above will have one degree of freedom. Along any of three lines on the phase diagram when one variable (pressure or temperature) is specified, the other is fixed automatically.

(3) Triple Point

The *three* boundary lines enclosing the *three* areas on the phase diagram intersect at a common point called the *Triple point*. A **triple point shows the conditions under which all the three phases (solid, liquid, vapour) can coexist in equilibrium**. Thus the system at the triple point may be represented as :



Applying the phase rule equation, we have

$$F = C - P + 2 = 1 - 3 + 2 = 0$$

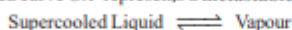
which predicts that the system has no degree of freedom.

At the triple point both pressure and temperature on the diagram are fixed and, therefore, the system is nonvariant. This implies that if we try to change temperature or pressure, the equilibrium will be disturbed. For example, if we lower the pressure on the system, all the liquid will vaporise, leaving only two phases.

In case of water system, the temperature and pressure for the triple point are 0.0076° and 4.58 mm Hg respectively.

Metastable Equilibrium

The vapour pressure curve AO of the liquid phase terminates at O , when the liquid freezes (or solidifies). However by careful cooling of the liquid under conditions that crystals do not form, the curve AO can be extended to A' . This means that the liquid can be cooled far below the freezing point or 'supercooled' without separation of the crystals. The supercooled liquid is in an unstable condition. On the slightest disturbance as introduction of a seed crystal, the entire liquid solidifies rapidly. Thus the dashed curve OA' represents a **metastable equilibrium**,



This system at once reverts to the true stable system



under suitable conditions. It is noteworthy that the dashed curve of the metastable liquid lies above the normal sublimation curve (BO). This implies that **the vapour pressure of the metastable phase is always higher than that of the stable phase at the same temperature.**

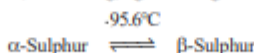
POLYMORPHISM

The occurrence of the same substance in more than one crystalline forms is known as Polymorphism. This phenomenon is shown by both elements and compounds. In the case of elements the term **allotropy** is often used. The individual crystalline forms of an element are referred to as **polymorphs** or **allotropes**. Rhombic and monoclinic sulphur are two polymorphs or allotropes of sulphur. The polymorphic or allotropic forms of an element have distinct physical properties and constitute separate phases.

Allotropy can be divided into three types : Enantiotropy, Monotropy and Dynamic allotropy.

(1) Enantiotropy

In some cases one polymorphic form (or allotrope) can change into another at a definite temperature when the two forms have a common vapour pressure. This temperature is known as the **transition temperature**. One form is stable above this temperature and the other form below it. **When the change of one form to the other at the transition temperature is reversible, the phenomenon is called enantiotropy and the polymorphic forms enantiotropes.** For example, rhombic sulphur (α -Sulphur) on heating changes to monoclinic sulphur (β -Sulphur) at 95.6°C (transition temperature). Also, monoclinic sulphur, on cooling, again changes to rhombic sulphur at 95.6°C . That is,



Thus α -Sulphur and β -Sulphur are enantiotropic.

(2) Monotropy

It occurs when one form is stable and the other metastable. **The metastable changes to the stable form at all temperatures and the change is not reversible.** Thus there is no transition temperature as the vapour pressures are never equal. This type of polymorphism is exhibited by phosphorus,



Some powdered rhombic sulphur is placed in the glass bulb and liquid paraffin (an inert liquid) is introduced above the sulphur. The apparatus is then immersed in a heating water bath, the temperature of which is raised. The scale reading and the temperature is recorded every minute. A plot of liquid level in the capillary against temperature gives a curve as in Fig. 19.2 (b). On cooling of the dilatometer, reverse changes take place but due to thermal lag the curve assumes the form shown in the figure. The transition temperature is taken as the mean of the respective temperatures where expansion starts (T_1) and contraction begins (T_2).

(3) Solubility change

Two forms of the same substance have different solubilities but at the transition point they have identical solubility. Thus if solubility-temperature graph is plotted for the two forms, it is found to consist of two parts with a sharp break. While one part represents the solubility curve for one form, the second part represents that for the other. **At the meeting point of the two curves, the solubility of the two forms is the same and it indicates the transition temperature.** For example, in the diagram for the system sodium sulphate-water, the solubility curves of Na_2SO_4 (rhombic) and $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ meet at 32.2°C . Thus 32.2°C is the transition temperature where $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ changes to Na_2SO_4 .

(4) Cooling curve method

There is often an evolution or absorption of heat when one form passes to the other. Suppose that form *A* is converted into form *B* on heating. Now let *B* be allowed to cool and a curve obtained by plotting the temperature against the time. **The otherwise steady curve has a distinct break at a temperature corresponding to the transition point because here heat is evolved from *B*.**

This method is suitable for determining the transition temperature between different hydrates of a salt or between a hydrate and an anhydrous salt (e.g., $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ to Na_2SO_4), or for different forms of a metal.

THE WATER SYSTEM

Under normal conditions the system 'water' is a **three-phase, one-component system**. The three phases involved are *liquid water*, *ice*, *water vapour*. All these phases can be represented by one chemical entity H_2O and hence one component of the system. The number of phases which can exist in equilibrium any time depends on the conditions of temperature and pressure. The phase diagram or *PT*-graph of the system/ water/ice/vapour is shown in Fig. 19.3. The **salient features of the phase diagram** are listed below.

- (1) The Curves *OA*, *OB*, *OC*
- (2) The Triple Point *O*
- (3) The Areas *AOC*, *AOB*, *BOC*

Let us proceed to discuss the significance of each of these features.

(1) The Curves *OA*, *OB*, *OC*

These three curves meet at the point *O* and divide the diagram into three regions or areas.

Curve *OA*, the Vapour Pressure curve of Water. It represents the vapour pressure of liquid water at different temperatures. The two phases *water* and *water vapour* coexist in equilibrium along this curve. The curve *OA* terminates at *A*, the critical point (218 atm , temp. 374°C) when the liquid and vapour are indistinguishable from each other and there is left one phase only. When the vapour pressure is equal to one-atmosphere, the corresponding temperature, as indicated on the phase diagram is the boiling point (100°C) of water.

Curve *OB*, the Sublimation curve of Ice. It shows the vapour pressure of solid ice at different temperatures. The two phases *solid ice* and *vapour* coexist in equilibrium along this curve. At the lower limit the curve *OB* terminates at absolute zero (-273°C) where no vapour exists.

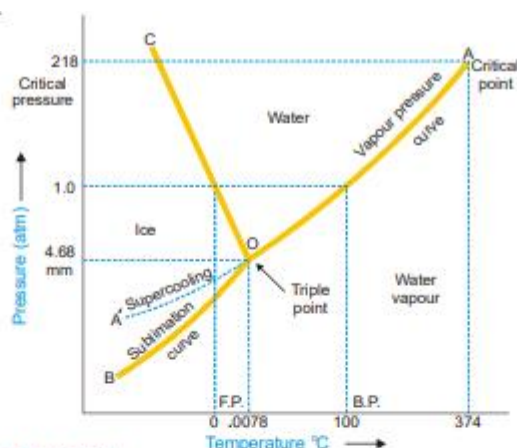


Figure 19.3
The phase diagram of the system 'Water'.

Curve OC, the Fusion curve of Ice. It depicts the effect of pressure on the melting point of ice. Here *ice* and *water* coexist in equilibrium. The fact that *OC* slopes to the left indicates that the melting point of ice decreases with increase of pressure. Since ice melts with decrease in volume by Le Chatelier's principle the melting point is lowered by an increase of pressure. It may be noted that the 1.0 atmosphere line meets the fusion curve at 0°C which is the normal melting point of ice.

Along the curves *OA*, *OB*, *OC* there are two phases in equilibrium and one component. Therefore,

$$F = C - P + 2 = 1 - 2 + 2 = 1$$

Hence each two-phase system :

water/water vapour represented by *OA*

ice/water vapour represented by *OB*

ice/water represented by *OC*

has one degree of freedom i.e., is **monovariant**.

(2) The Triple point 'O'

The curves *OA*, *OB* and *OC* meet at the triple point 'O' where all the three phases *liquid water/ice/vapour* are in equilibrium. This occurs at 0.0076°C and vapour pressure 4.58 mm Hg. Since there are three phases and one component, we have

$$F = C - P + 2 = 1 - 3 + 2 = 0$$

i.e., the system at the triple point is **nonvariant**. Thus if either pressure or temperature is changed, the three phases would not exist and one of the phases would disappear.

(3) Area *AOC*, *AOB*, *BOC*

The areas or regions between the curves show the conditions of temperature and pressure under which a single phase—**ice, water or vapour** is capable of stable existence. Thus,

Area *AOC* represents conditions for the one-phase system **water**.

Area *AOB* represents conditions for the one-phase system **water vapour**.

Area *BOC* represents conditions for the one-phase system **ice**.

In all the three areas there being one-phase and one-component, we have

$$F = C - P + 2 = 1 - 1 + 2 = 2$$

Thus each system *water*, *water vapour*, or *ice* has 2 degrees of freedom i.e., the system is **bivariant**.

(4) Metastable system : Supercooled Water/Vapour System

The vapour pressure curve of water *AO* can be continued past the triple point as shown by the dashed line *OA'*. That is, water can be *supercooled* by carefully eliminating solid particles. The supercooled water/vapour system is metastable (unstable). It at once reverts to the stable system ice/vapour on the slightest disturbance or introducing a crystal of ice.

CHAPTER

CONTENTS

WAYS OF EXPRESSING
CONCENTRATION

MOLARITY

MOLALITY

NORMALITY

SOLUTIONS OF GASES IN GASES

HENRY'S LAW

SOLUTIONS OF LIQUIDS IN
LIQUIDS

SOLUBILITY OF COMPLETELY
MISCIBLE LIQUIDS

SOLUBILITY OF PARTIALLY
MISCIBLE LIQUIDS

PHENOL-WATER SYSTEM

TRIMETHYLAMINE-WATER
SYSTEM

NICOTINE-WATER SYSTEM

VAPOUR PRESSURES OF LIQUID-
LIQUID SOLUTIONS

AZEOTROPE

THEORY OF FRACTIONAL
DISTILLATION

STEAM DISTILLATION

SOLUTIONS OF SOLIDS IN LIQUIDS

SOLUBILITY-EQUILIBRIUM
CONCEPT

DETERMINATION OF SOLUBILITY



A solution is a **homogeneous mixture** of two or more substances on **molecular level**. The constituent of the mixture present in a **smaller amount** is called the **Solute** and the one present in a larger amount is called the **Solvent**. For example, when a smaller amount of sugar (solute) is mixed with water (solvent), a homogeneous solution in water is obtained. In this solution, sugar molecules are uniformly dispersed in molecules of water. Similarly, a solution of salt ($\text{Na}^+ \text{Cl}^-$) in water consists of ions of salt (Na^+, Cl^-) dispersed in water.

CONCENTRATION OF A SOLUTION

The concentration of a solution is defined as : **the amount of solute present in a given amount of solution.**

Concentration is generally expressed as the quantity of solute in a unit volume of solution.

$$\text{Concentration} = \frac{\text{Quantity of solute}}{\text{Volume of solution}}$$

Figure 14.1
Molecular model of a solution.

TYPES OF SOLUTIONS

The common solutions that we come across are those where the solute is a solid and the solvent is a liquid. In fact, substance in any three states of matter (solid, liquid, gas) can act as solute or solvent. Thus there are **seven types of solutions** whose examples are listed in Table 14.1.

State of Solute	State of Solvent	Example
1. Gas	Gas	Air
2. Gas	Liquid (Carbonated drinks)	Oxygen in water, CO ₂ in water
3. Gas	Solid	Adsorption of H ₂ by palladium
4. Liquid	Liquid	Alcohol in water
5. Liquid	Solid	Mercury in silver
6. Solid	Liquid	Sugar, Salt
7. Solid	Solid (Steel)	Metal alloys : Carbon in iron

Of the seven types of solutions mentioned in Table 14.1, we will discuss only the important ones in detail.

WAYS OF EXPRESSING CONCENTRATION

There are several ways of expressing concentration of a solution :

- Per cent by weight
- Mole fraction
- Molarity
- Molality
- Normality

Per cent by Weight

It is the **weight of the solute as a per cent of the total weight of the solution**. That is,

$$\% \text{ by weight of solute} = \frac{\text{Wt. of solute}}{\text{Wt. of solution}} \times 100$$

What are molarity, molality, and normality? Explain with an example.

Solution

Molarity

- Molarity of solution is a weight by volume relationship to represent its strength and is defined as 'the number of moles of solute dissolved in one liter of solution.

Formula of molarity

$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution in litre}}$$

Example of molarity

- 1.25 mol L⁻¹ or (1.25 M) solution of KOH means that 1.25 mol of KOH has been dissolved in one liter (or one cubic decimetre).

Molality

- Molality (m) is defined as the number of moles of the solute per kilogram (kg) of the solvent.

Formula for molality

$$\text{Molality} = \frac{\text{Moles of solute}}{\text{Mass of solvent in kg}}$$

Example of molality

- 1 mol kg⁻¹ (or 1 m) solution of KCl means that 1 mol (74.5 g) of KCl is dissolved in 1 kg of water.

Normality

- It is the measure of concentration equal to the gram equivalent mass of a substance per liter of solution.

MOLALITY

Molality of a solution (symbol m) is defined as **the number of moles of solute per kilogram of solvent** :

$$\text{Molality } (m) = \frac{\text{Moles of solute}}{\text{Mass of solvent in kilograms}}$$

A solution obtained by dissolving one mole of the solute in 1000 g of solvent is called **one molal** or **1m solution**.

Notice the difference between molality and molarity. **Molality is defined in terms of mass of solvent while molarity is defined in terms of volume of solution.**

SOLVED PROBLEM. What is the molality of a solution prepared by dissolving 5.0 g of toluene (C_7H_8) in 225 g of benzene (C_6H_6) ?

SOLUTION

Calculation of number of moles of solute :

$$\text{Molecular mass of toluene} = 12 \times 7 + 1 \times 8 = 92$$

$$\text{No. of moles 5 g of toluene} = \frac{5}{92} = 0.0543$$

$$\text{Mass of solvent in kg} = \frac{225 \text{ g}}{1000} = 0.225 \text{ kg}$$

$$\text{Molality} = \frac{\text{No. of moles of solute}}{\text{Mass of solvent in kg}} = \frac{0.0543}{0.225}$$

For example, if a solution of NaCl contains 50 per cent NaCl by weight, it has 50 g of NaCl for 100 g of solution.

SOLVED PROBLEM. What is the per cent by weight of NaCl if 1.75 g of NaCl is dissolved in 5.85 g of water.

SOLUTION

$$\text{Wt. of solute (NaCl)} = 1.75 \text{ g}$$

$$\text{Wt. of solvent (H}_2\text{O)} = 5.85 \text{ g}$$

$$\therefore \text{Wt. of solution} = 1.75 + 5.85 = 7.60 \text{ g}$$

$$\begin{aligned} \text{Hence concentration of NaCl \% by weight} \\ &= \frac{1.75}{7.60} \times 100 \\ &= 23.0 \end{aligned}$$

MOLE FRACTION

A simple solution is made of two substances : one is the solute and the other solvent. Mole fraction, X , of solute is defined as **the ratio of the number of moles of solute and the total number of moles of solute and solvent**. Thus,

$$X_{\text{solute}} = \frac{\text{Moles of solute}}{\text{Moles of solute} + \text{Moles of solvent}}$$

If n represents moles of solute and N number of moles of solvent,

$$X_{\text{solute}} = \frac{n}{n + N}$$

Notice that mole fraction of solvent would be

$$X_{\text{solvent}} = \frac{N}{n + N}$$

Mole fraction is unitless and

$$X_{\text{solute}} + X_{\text{solvent}} = 1$$

SOLVED PROBLEM. Calculate the mole fraction of HCl in a solution of hydrochloric acid in water, containing 36 per cent HCl by weight

SOLUTION

The solution contains 36 g of HCl and 64 g of H_2O

$$\begin{aligned} \text{Number of Moles of HCl} &= (36 \text{ g HCl}) \left(\frac{1 \text{ mol HCl}}{36.5 \text{ g HCl}} \right) \\ &= 0.99 \end{aligned}$$

$$\begin{aligned} \text{Number of Moles of H}_2\text{O} &= (64 \text{ g H}_2\text{O}) \left(\frac{1 \text{ mol H}_2\text{O}}{18 \text{ g H}_2\text{O}} \right) \\ &= 3.55 \end{aligned}$$

$$\begin{aligned} X_{\text{HCl}} &= \frac{\text{moles of HCl}}{\text{moles of HCl} + \text{moles of H}_2\text{O}} \\ &= \frac{0.99}{3.55 + 0.99} = 0.218 \end{aligned}$$

Do it.

1. Calculate the mole fraction of H_2SO_4 in solution of H_2SO_4 in water having 36% of the acid by weight.
2. .02 kg of NaCl is dissolved in 1000 g of water. If the density of the solution is 0.809 g/mL, calculate the molarity, molality and mole fraction of the solute.

$$M = x \text{ g} \times \frac{1 \text{ mol}}{MW \text{ g}} \times \frac{1}{V} \quad \dots(2)$$

From expression (1) can also be found the amount of solute in grams if molarity is given.

SOLVED PROBLEM 1. What is the molarity of a solution prepared by dissolving 75.5 g of pure KOH in 540 ml of solution.

SOLUTION

$$\begin{aligned} \text{Molecular mass of KOH} &= 39.1 + 16.0 + 1.0 \\ &= 56.1 \end{aligned}$$

Calculation of moles of KOH:

$$75.5 \text{ g KOH} \times \frac{1 \text{ mol}}{56.1 \text{ g}} = 1.35 \text{ mol}$$

Calculation of volume in litres :

$$540 \text{ ml} \times \frac{1 \text{ litre}}{1000 \text{ ml}} = 0.540 \text{ litres}$$

532 **14** PHYSICAL CHEMISTRY

Calculation of Molarity :

$$\begin{aligned} M &= \frac{n}{V} \\ &= \frac{1.35 \text{ mol}}{0.540 \text{ litre}} = 2.50 \text{ M} \end{aligned}$$

Thus the solution is **2.50 M**.

SOLVED PROBLEM 2. What weight of HCl is present in 155 ml of a 0.540 M solution ?

SOLUTION

$$M = 0.540$$

$$V \text{ in litres} = \frac{155}{1000} = 0.155 \text{ litre}$$

$$n = M \times V$$

$$= 0.540 \text{ mol/litre} \times 0.155 \text{ litre}$$

$$= 0.0837 \text{ mol of HCl}$$

$$0.0837 \text{ mol of HCl} \times \frac{36.5 \text{ g}}{\text{mol HCl}} = 3.06 \text{ g HCl}$$

Thus **3.06 g of HCl** is present in 155 ml of the given solution.

MOLALITY

Molality of a solution (symbol m) is defined as the number of moles of solute per kilogram of

Do it.

Calculate the amount of HNO_3 is present in 150 mL of a 0.50 M solution

The relationship between pressure and solubility of a gas in a particular solvent was investigated by William Henry (1803). He gave a generalisation which is known as **Henry's Law**. It may be stated as : **for a gas in contact with a solvent at constant temperature, concentration of the gas that dissolves in the solvent is directly proportional to the pressure of the gas.**

SOLUTIONS 535

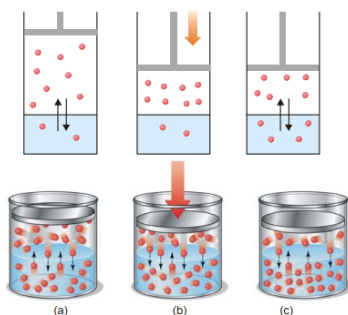


Figure 14.2
(a) Equilibrium exists between the gas molecules and those in solution; (b) When pressure is applied, gas volume decreases and concentration in gas phase increases; (c) More gas molecules pass into solution to reestablish equilibrium whereby the pressure above the gas is decreased.

Mathematically, Henry's Law may be expressed as

$$C \propto P$$

equilibrium whereby the pressure above the gas is decreased.

Mathematically, Henry's Law may be expressed as

$$C \propto P$$

or

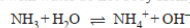
$$C = k P$$

where P = pressure of the gas; C = concentration of the gas in solution; and k = proportionality constant known as **Henry's Law Constant**. The value of k depends on the nature of the gas and solvent, and the units of P and C used.

Limitations of Henry's Law

It applies closely to gases with nearly ideal behaviour.

- (1) at moderate temperature and pressure.
- (2) if the solubility of the gas in the solvent is low.
- (3) the gas does not react with the solvent to form a new species. Thus ammonia (or HCl) which react with water do not obey Henry's Law.



- (4) the gas does not associate or dissociate on dissolving in the solvent.

SOLVED PROBLEM 1. The solubility of pure oxygen in water at 20°C and 1.00 atmosphere pressure is 1.38×10^{-3} mole/litre. Calculate the concentration of oxygen at 20°C and partial pressure of 0.21 atmosphere.

SOLUTION

Calculation of k :

$$C = k P$$

or

$$k = \frac{C}{P}$$